



NewBear Computing Store Ltd



HEAD OFFICE: 40 BARTHOLOMEW ST. NEWBURY BERKS. RG14 5LL Tel: (0635) 30505 Telex: 848507NCS

79 - 09

6809 C.P.U. Board.

Design Note 81

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79-09 CPU BOARD

1. INTRODUCTION
2. BOARD CHARACTERISTICS
3. CIRCUIT DESCRIPTION
4. CONSTRUCTION
5. TESTING

- Figs; 1. Schematic
2. Component side view of board
 3. Component list
 4. Connector wiring
 5. Optional strapping
 6. Memory map

Appendices;

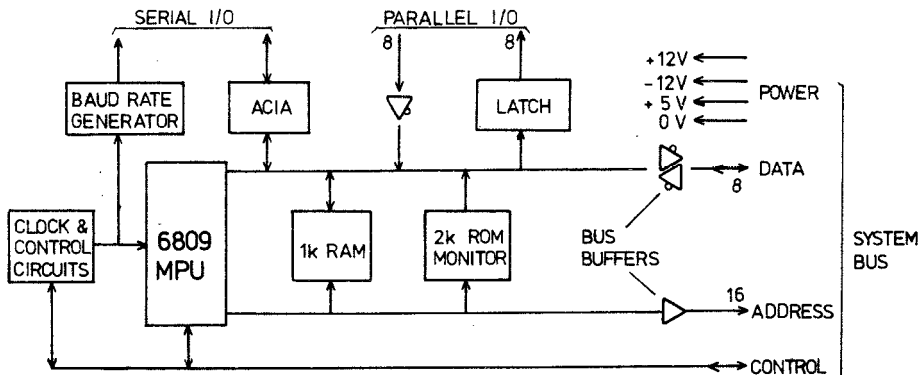
1. STWPC SBUG-E Monitor
2. Using 77-68 boards

1. INTRODUCTION

This board carries a 6809 microprocessor, 1k bytes of RAM, a 1 or 2k byte ROM Monitor, and parallel and serial I/O ports.

It may be used as a stand-alone microcomputer board connected to a Teletype or similar terminal, or as the heart of a powerful microcomputer system incorporating other boards from the 77-68 range.

Addresses of on-board ROM, RAM & I/O are compatible with the SWTPC 6809 system, allowing the use of widely available software.



79-09 BLOCK DIAGRAM

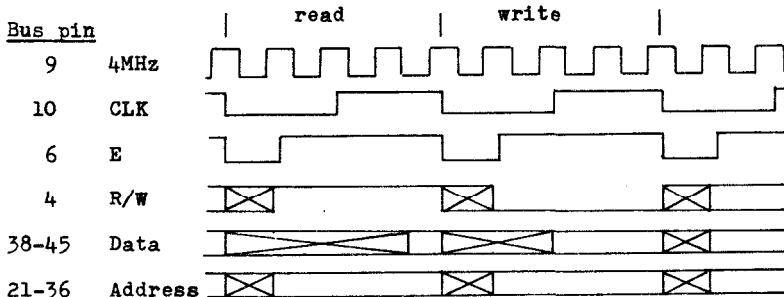
2. BOARD CHARACTERISTICS

Construction	; 8.0" x 8.0" single sided PCB with gold plated edge connector.
Bus	; 77-68 system compatible.
CPU	; 6809 8-bit microprocessor; selectable 1 or 2uS cycle time.
RAM	; 1k bytes on board; extendable to 56k bytes off board.
ROM	; 1 or 2k byte ROM or EPROM monitor.
Parallel I/O	; 8 bits input, 8 bits output. LS TTL levels.
Serial I/O	; Asynchronous (Stop-Start) RS232 or TTL levels. Crystal controlled baud rates selectable 75 - 19200 baud.
Power	; Requires +5V @ 0.6A nom. +12V @ 50mA nom. -12V @ 50mA nom.

3. CIRCUIT DESCRIPTION

MPU Clocks; The output of the 4MHz crystal controlled oscillator is fed to edge connector pin 9 and also to the 4-stage binary divider (part of X22). By strapping 'l' to 'm' or 'k', either 2MHz or 4MHz can be applied to the MPU clock input (EXTAL) via buffer X20.

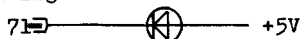
This clock is further divided within X15 to give two timing signals Q and E' (X15 pins 35 and 34). E' is buffered and fed to edge connector pin 10, being compatible with the 77-68 'CLK' signal. By OR'ing Q and E' in X4, a signal is generated at edge connector pin 6 which is compatible with the 77-68 Bus E signal. The relationship between these, and other Bus control signals, is shown below for a μ S MPU cycle time;



Bus Control Signals; The HALT and RESET inputs (board pins 72,73) should normally be high (pulled to +5V by the 4k7 resistors), but connected via switches to OV to Halt or Reset the MPU. The HOLD input (board pin 8) is normally high, but may be pulled to OV for a period not exceeding 10 μ S during a MPU read or write operation to cater for unusually slow memory.

The 77-68 bus has two interrupt lines; IRQ & NMI (pins 11 & 12), however as the 6809 MPU has three interrupt inputs (IRQ, NMI and the Fast Interrupt Request FIRQ), strapping points w,x,y,aa,ab are provided to allow the user to choose any two. R3,4,5 normally hold these lines at +5V; they are pulled down to OV by external circuitry to cause an interrupt.

The R/W signal (bus pin 4) is high for a read from memory to the MPU, and low for a write from the MPU to memory. Edge connector pin 71 (RUNLMP) should be connected to +5V via a LED, which will light when the MPU is running;



Data and Address Buffers; X7 and X11 buffer the 6809 address lines A0-A15 for feeding onto the bus, and are permanently enabled. X12 is a bi-directional tri-state buffer used to transfer data between the system bus and the 79-09 board. It is enabled by BUFEE being low; the direction of data transmission then being controlled by the R/W signal applied to X12 pin 1. When on-board memory or I/O ports are accessed, BUFEE goes high to disable X12 and isolate the 79-09 board from the system data bus.

RAM and ROM; X9,10 form the 1k x 8 bit on board static RAM; which is enabled when RAME is low.

A 1 or 2k byte ROM or EPROM may be fitted in the X8 position to hold the system monitor program; connections being made to strapping points a - j according to the type of IC used (see Fig 5). X8 is enabled when ROME goes low.

Parallel I/O; X16 is an 8 bit latch which can be written into by a MPU write operation to address E000. Its outputs are taken to board edge connector pins 57-64. Up to eight parallel inputs can be presented to board connector pins 49-56 (note the 4k7 pull-up resistors), and the state of these inputs examined by a MPU read operation from address E000; which enables the tri-state buffer X17 to transfer the information onto the system data bus.

Serial I/O; ACIA X13 converts between the 8-bit parallel form used by the MPU, and the serial stop-start format required by the terminal. Two Serial Out ports are provided; one buffered by amplifier X14 to RS232 levels, the other is at normal TTL levels. The single Serial In port can accept either RS232 or TTL level signals.

X22 and X23 divide down the 4MHz clock according to connections of strapping points k,n-s,R and Ra-Re (see Fig 5) to provide a baud rate clock on edge connector pin 70 (BD). This, or clock signals from external sources, should be connected to the TX and RX clock inputs (edge connector pins 67 & 68) for correct timing of the ACIA operations.

Address decoding; The One out of Eight decoder X6, in conjunction with the surrounding gates (parts of X2,3,4,5) generate enable signals for the ROM, RAM, I/O ports and Data Bus Buffer according to the address present on lines A0-A15, and also the state of the R/W and E signals;

A:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Enabled
	1	1	1	1	1	A	A	A	A	A	A	A	A	A	A	A	ROM
	1	1	1	0	0								0	1		R	ACIA
	1	1	1	0	0								0	0			Parallel ports
	1	1	0	1			A	A	A	A	A	A	A	A	A	A	RAM

Notes; A = address of particular ROM or RAM location
R = ACIA Register select bit

4. CONSTRUCTION

Before fitting any components, examine the track side of the board carefully for unetched copper 'bridges', unintentional breaks in tracks, or undrilled holes.

The close spacing of the tracks in some areas of the board means that great care has to be taken to avoid accidental shorts between tracks. The constructor should therefore take care to make good soldered joints using the minimum amount of solder whilst ensuring that a joint is made all round the component lead or wire end. The use of a small, clean iron is essential.

IC sockets should be fitted in X8,9,10,13 & 15 positions at least; whether sockets are used for the other IC's is a matter for personal preference.

The best order of construction is;

First decide on which strapping options are to be used (see Fig 5), then fit all straps, using sleeved wire as appropriate.

Then fit the resistors followed by the IC sockets then the ICs which are not to be fitted in sockets.

Finally, fit the remaining components.

After assembly, clean the track side of the board thoroughly with a small stiff brush, and examine it carefully for short circuits caused by bent leads, excess solder on joints, solder splashes etc. Then re-check that the correct components have been fitted and that all ICs, tantalum capacitors and the diode and transistor have been fitted the correct way around.

5. TESTING

First, carefully check with an ohm-meter for short-circuits between adjacent tracks (especially in the crowded centre area of the board) and also between 0V and +5V, +12V & -12V lines. Note that some reading is to be expected due to the internal resistances of the IC's. This stage is most important as it can reveal faults which would otherwise be extremely difficult to locate.

Apply power to the board and leave it for a few minutes while monitoring the power supply voltages and checking for signs of overheating. Note that if a 2708 type of EPROM is used for X8 it is important to switch all power supply lines on and off simultaneously, as data may be lost if the positive supplies are left connected without the negative supply.

With a suitable terminal connected to the board (check for correct baud rates), and the HALT switch off, momentarily operating the RESET switch should set the system running and the monitor prompt character will be printed.

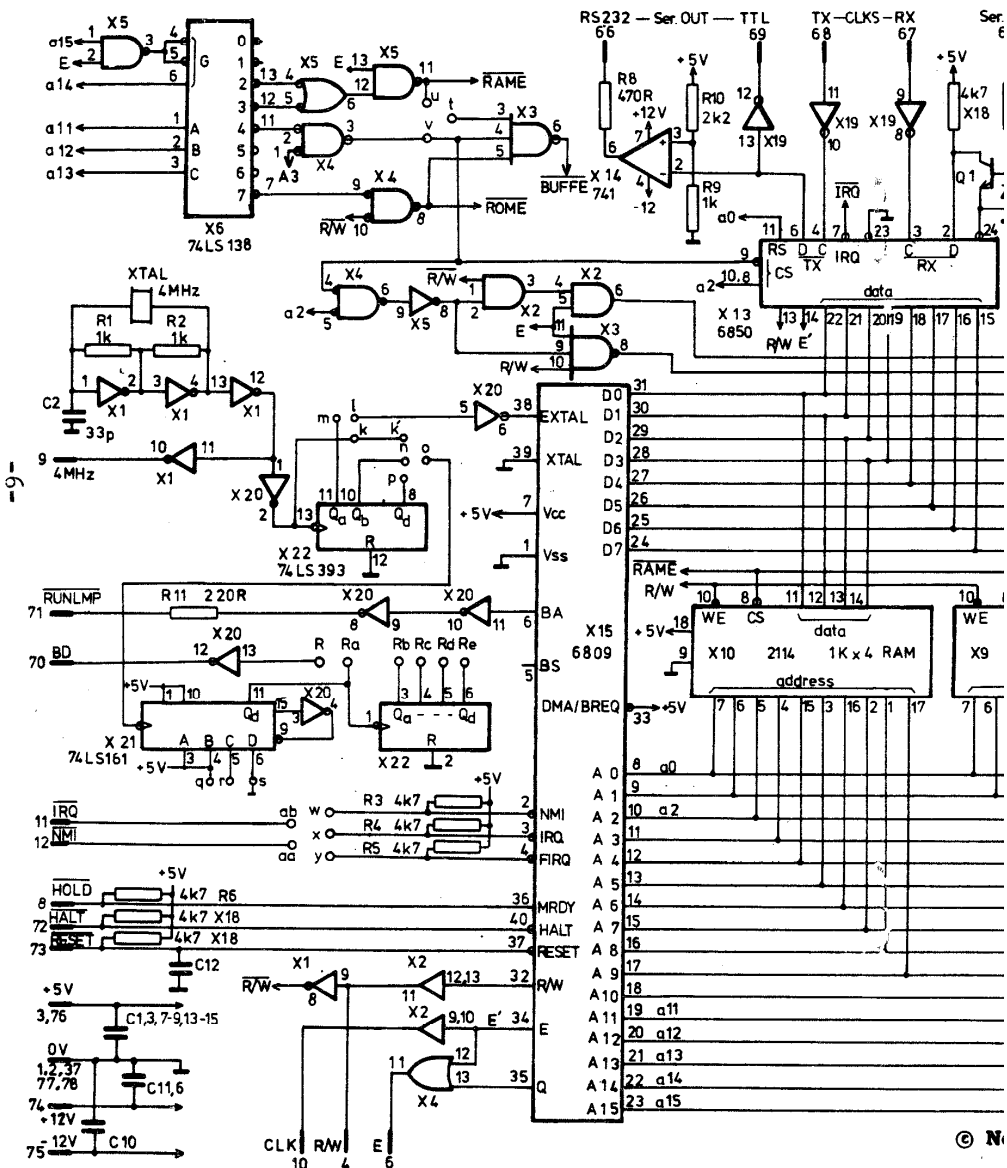
If all is not well, then the following tests can be used to find which area of the board is causing the problem;

4 MHz oscillator; If you have a high speed CRO, then a 4MHz square wave should be seen on X1 pins 10 & 12, and X20 pin 2. X15 pin 38 should have either a 4MHz or a 2MHz square waveform depending on the strapping to 1. If you haven't got a suitable CRO, then a moving coil DC voltmeter should show about +2V at these points, and this should either rise to about 3.5V or fall to about 0V if the oscillator is stopped by connecting X1 pin 1 to 0V.

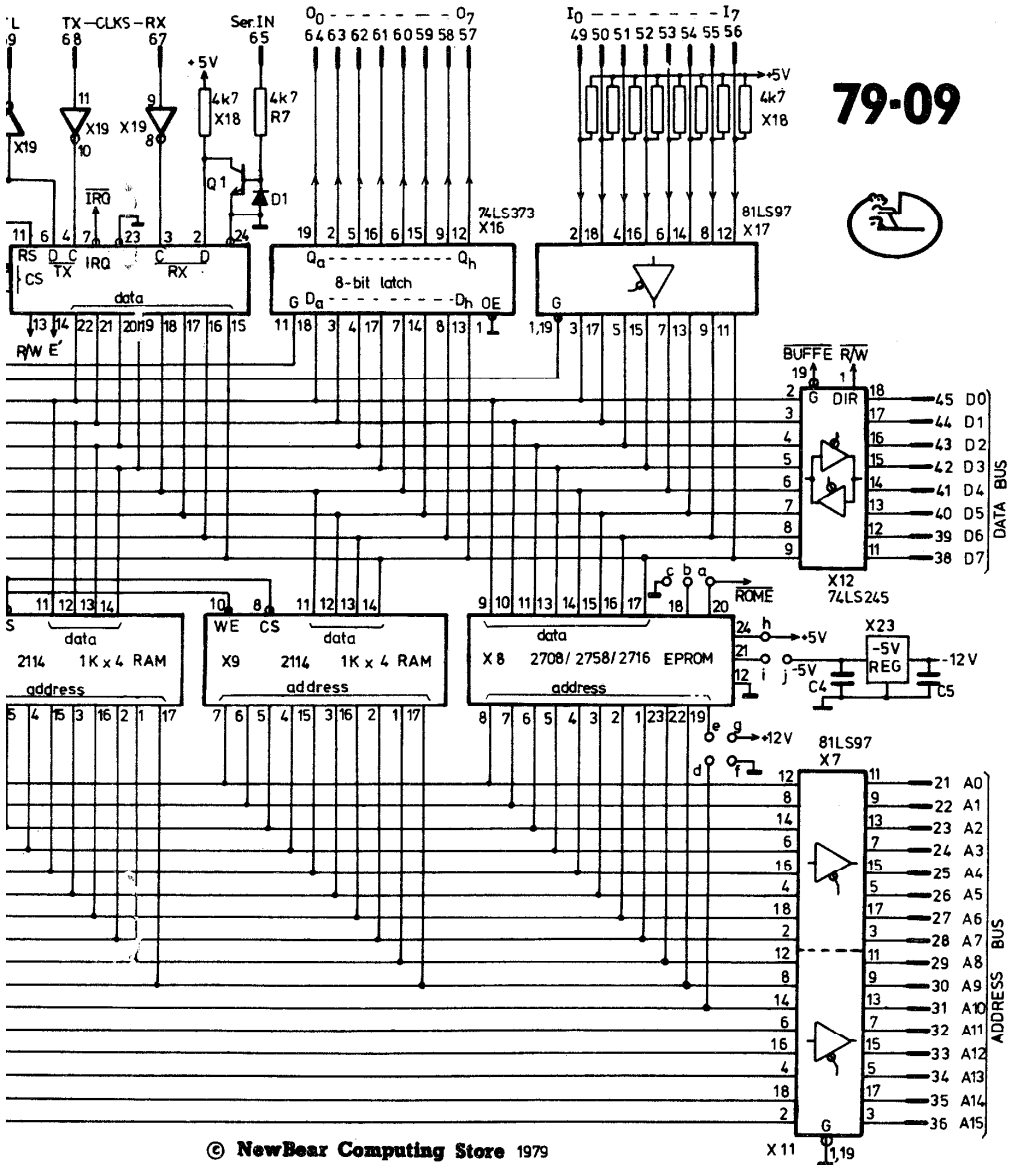
E and CLK; If the signal on X15 pin 38 is OK, then X15 pins 34 & 35 should each give a 1MHz (or 0.5MHz) square wave, as should board pin 10. Board connector pin 6 should show an asymmetric waveform (see section 3), and measure about +3V on a moving coil meter.

Baud Rate generator; Board connector pin 70 (BD) should have a signal at 16 times the selected baud rate, as should X13 clock inputs (X13 pins 3 & 4).

Serial data buffers; When nothing is being sent or received from the terminal, X13 pin 6 should be high, X19 pin 12 should be low, and RS232 Out (board pin 66) should be about -10V, while Serial In (board pin 65) should be 0V or negative, and X13 pin 2 high.



79-09





79-09 CPU BOARD

Components

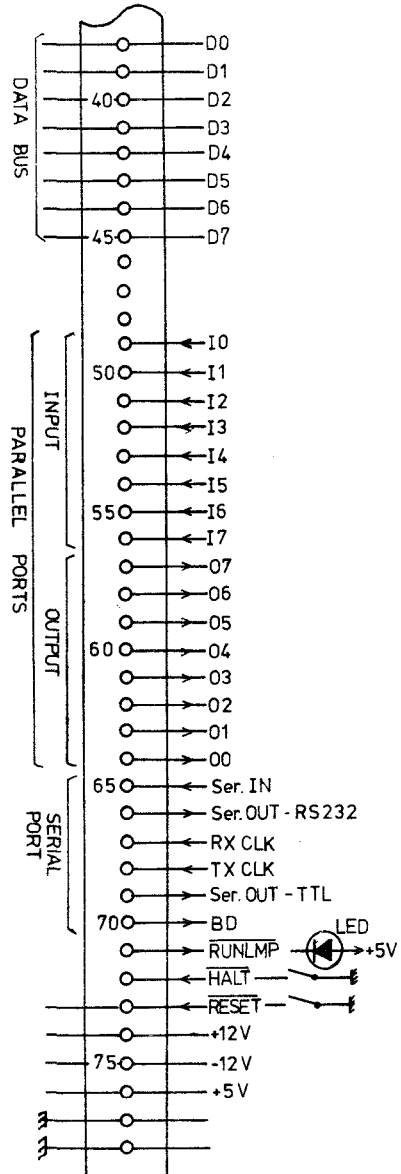
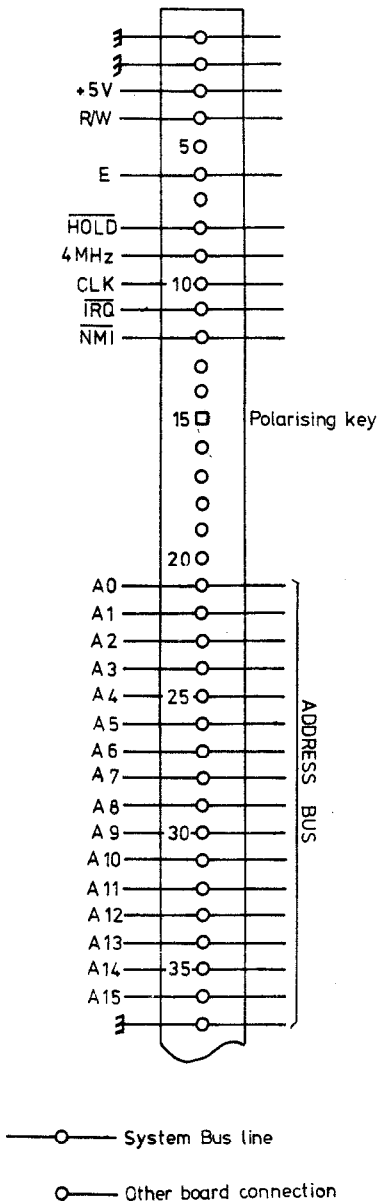
X1,19,20	74LS04	; hex inverter
X2	74LS08	; quad 2 i/p AND gate
X3	74LS10	; triple 3 i/p NAND gate
X4	74LS32	; quad 2 i/p OR gate
X5	74LS00	; quad 2 i/p NAND gate
X6	74LS138	; 1/8 decoder
X7,11,17	81LS97	
	or 81LS95	; octal tri-state buffer
X8	see text	; EPROM or ROM
X9,10	2114	; 1k x 4 static RAM
X12	74LS245	; 8 bit bi-directional tri-state buffer
X13	6850	; ACIA
X14	741	; op amp (8 pin DIL)
X15	6809	; MPU
X16	74LS373	; octal latch
X18		; resistor network, 13 x 4k7 RS stock no. 140-)41
X21	74LS161	; programmable counter
X22	74LS393	; dual 4 bit binary counter
X23	79L05	; -5V regulator
Q1	BC107	; npn transistor
D1	1N4148	; diode
R1,2,9	1k	; resistor $\frac{1}{4}$ W
R3-7	4k7	; resistor $\frac{1}{4}$ W
R8	470R	; resistor $\frac{1}{4}$ W
R10	2k2	; resistor $\frac{1}{4}$ W
R11	220R	; resistor $\frac{1}{4}$ W
C1,5,7,10,11,13,14,15	4u7 25V	; tantalum bead capacitor
C2	33p	; ceramic capacitor
C3,4,6,8,9,12	10n 30V	; ceramic capacitor
XTAL	4MHz	
1 off 40-pin DIL socket		; for X15
2 off 24-pin DIL socket		; for X8,13
2 off 18-pin DIL socket		; for X9,10

79-09 printed circuit board

Edge connector; 77 way + 1 polarising position
(78 positions total) 0.1" pitch single sided.

Fig. 3 COMPONENT LIST

79-09 : CONNECTOR WIRING



EPROM type (strapping points a - j)

2708 b-c e-g i-j
2758 b-a e-f h-i
2716 b-a e-d h-i (+5V 2716 or SBUG-E)

MPU CYCLE TIME (strapping points l - m)

1uS l-k
2uS l-m

BAUD RATE OUTPUT BD (strapping points n - s, R, Ra - Re)

Baud rate for R connected to					Other straps
Ra	Rb	Rc	Rd	Re	
19200	9600	4800	2400	1200	o-k' r-s
4800	2400	1200	600	300	o-n r-s
1200	600	300	150	75	o-p r-s
				110	o-p r-q

RAM SELECT (strapping points u,v,t)

t is normally connected to u; connecting t to v disables the on-board 1k byte RAM allowing address space D000 - DFFF to be used by external memory or peripherals.

INTERRUPTS (strapping points w,x,y,aa,ab)

ab is normally connected to x, and aa to w (y unconnected)

Fig. 5; OPTIONAL STRAPPING

FFFF	2 k byte ROM Monitor
F800	
F7FF	
F400	
F3FF	PIO boards
F000	
EFFF	
EC00	
	VDU board
E7FF	
E000	
DFFF	
DC00	1 k byte RAM
DBFF	
D000	
	Echoes of 1 k byte RAM
	Free for external RAM
0000	

Notes; If a 1k byte Monitor ROM is fitted in X8 position, it will appear at FC00-FFFF and also at F800-FBFF.

VDU board modified as appendix 2.

Parallel port echoes throughout address space E000 - E7F3 in all addresses with last digit 0,1,2 or 3. e.g. E003, E641 etc.

Serial port echoes throughout address space E004 - E7F7 in all addresses ending in 4,5,6 or 7. e.g. E006, E444 etc.

External peripheral ports added to the system should therefore use only those addresses in the range E000 - E7FF which end in 8,9,A,B,C,D,E or F.

Fig. 6; MEMORY MAP

Appendix 1

NOTES ON THE SWTPC SBUG-E 6809 ROM MONITOR

Although SBUG-E was designed for use with SWTPC's MP-09 processor board, and has facilities not required by 79-09, it does however function as a useful Monitor for 79-09 systems using a Teletype or similar terminal connected to the on-board ACIA. Its chief features are listed below;

Monitor Commands

CTRL/A	Examine / alter accumulator A
CTRL/B	Examine / alter accumulator B
CTRL/C	Examine / alter condition code register
CTRL/D	Examine / alter direct page register
CTRL/P	Examine / alter program counter
CTRL/U	Examine / alter user stack pointer
CTRL/X	Examine / alter X register
CTRL/Y	Examine / alter Y register
B addr	Set breakpoint (up to 5 breakpoints may be set)
E addr - addr	Examine memory contents
G	Continue execution of user program
L	Load Mikbug format tape from terminal
M addr	Examine / alter memory byte
P addr - addr	Punch Mikbug format tape on terminal
Q addr - addr	Test memory area
R	Display registers
X	Remove breakpoints

RAM Used by SBUG-E

DF80 - DFFF

User Callable ROM Routines

ROM locations at the beginning of the Monitor (from F800) contain the addresses of monitor subroutines which may be called by the user program ; via an indirect JSR;

Addr	Name	
F800	MONITOR	Re-enter monitor
F802	NEXTCMD	Re-enter monitor and prompt for command
F804	INCH	Get input character from terminal in A reg, without echo.
F806	INCHE	Get character from terminal in A reg and echo it back to the terminal.
F808	INCHEK	Returns Z clear if a character is ready to be read from the receive half of the ACIA.
F80A	OUTCH	Output character in A reg to terminal
F80C	PDATA	Output a character string pointed to by X, ended with a '04' byte
F80E	PCRLF	Output CR, LF to terminal
F810	PSTRNG	Does PCRLF, then PDATA

Appendix 2

USING 77-68 BOARDS WITH THE 79-09 CPU

Because of slight differences between the 77-68 and 79-09 processors, some 77-68 boards will require slight modification to work with the 79-09 CPU;

32k Dynaram

This will only work if the 79-09 is set for a 2uS cycle time. Note also that the system bus line pin 9 now carries 4MHz instead of 5MHz, therefore the baud rate generation circuitry on the 32k Dynaram board will give incorrect frequency outputs.

4k RAM, ROM.A & PIO Boards

These can be used without modification.

77-68 VDU Board

This can be used by modifying it so that it occupies memory space E000 - EFFF. To do this cut the tracks going to edge connector pads 33 & 31 (A12 & A10), then re-connect them swapped over so that edge connector pad 33 (A12) goes to X19 pin 5, and edge connector pad 31 (A10) goes to X16 pin 9.

MON 1 and MON 2 Boards

It is not expected that these will be used with the 79-09.